

Mathematical Foundations of Linear, Mixed-Integer
and Convex Quadratic Optimization
The Suprema Optimization Solver: Theorems, Methods and References

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1 Notation and Conventions

Symbol	Meaning
$A \in \mathbb{R}^{m \times n}$	constraint matrix (rows $a_i^\top$, columns a_j)
b, c	right-hand side, linear cost
$Q \in \mathbb{R}^{n \times n}$	symmetric quadratic cost matrix
$l, u \in (\mathbb{R} \cup \{\pm\infty\})^n$	variable bounds
$\mathcal{I} \subseteq \{1, \dots, n\}$	index set of integer variables; $\mathcal{I}_{\text{bin}} \subseteq \mathcal{I}$ binaries
$y \in \mathbb{R}^m$	dual variables (row multipliers)
s, z	reduced costs / dual slack for variable bounds
$\mathcal{B}_s, \mathcal{N}$	basic and nonbasic index sets (simplex); $B = A_{\mathcal{B}_s}$
D_1, D_2	positive diagonal row and column scaling matrices
$X = \text{diag}(x), S = \text{diag}(s)$	diagonal matrices of iterates
e	all-ones vector
μ	barrier / duality-measure parameter
$\kappa(M)$	condition number $\ M\ \ M^{-1}\ $
proj_C	Euclidean projection onto a closed convex set C
ϵ_{mach}	unit roundoff of floating-point arithmetic

Conventions. All problems are minimizations (an LP maximization is converted by $c \rightarrow -c$; a QP maximization by $(Q, c) \rightarrow (-Q, -c)$, which is convex only if the original $Q \preceq 0$). Row senses are $\leq, \geq, =$; a ranged row is $r_i^{lo} \leq a_i^\top x \leq r_i^{hi}$. Norms are $\|\cdot\|_\infty$ unless stated.

2 Problem Formulations

Definition 2.1 (LP).

$$\min_{x \in \mathbb{R}^n} c^\top x \quad \text{s.t.} \quad Ax \in \{\leq, =, \geq\} b, \quad l \leq x \leq u.$$

Definition 2.2 (Convex QP).

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^\top Q x + c^\top x \quad \text{s.t.} \quad Ax \in \{\leq, =, \geq\} b, \quad l \leq x \leq u, \quad Q = Q^\top \succeq 0.$$

Definition 2.3 (MILP). The LP of Section 2 with the additional requirement $x_j \in \mathbb{Z}$ for $j \in \mathcal{I}$. Its *LP relaxation* drops integrality.

Definition 2.4 (Computational standard form). Adding a slack $w_i \geq 0$ (or $w_i \leq 0$) to each inequality row gives

$$\min c^\top x \quad \text{s.t.} \quad Ax + w = b, \quad l \leq x \leq u, \quad l_w \leq w \leq u_w,$$

where an equality row has $l_w = u_w = 0$ and an inequality has one finite bound at 0. Defining $\hat{A} = [A \ I]$, $\hat{x} = (x, w)$ gives $\hat{A}\hat{x} = b$ with box bounds.

Proposition 2.5 (Equivalence of forms). *Every LP (QP) is equivalent to one in standard form $\min\{c^\top x : Ax = b, x \geq 0\}$ via slacks, variable shifts $x \rightarrow x - l$, splitting of free variables $x = x^+ - x^-$, and explicit rows for finite upper bounds. Optimal values coincide and solutions map back explicitly [1, 2].*

Definition 2.6 (Problem classification). Let $q = [Q \neq 0]$ and $\iota = [\mathcal{I} \neq \emptyset]$. Then: $(q, \iota) = (0, 0)$ is an LP; $(0, 1)$ a MILP; $(1, 0)$ a QP; $(1, 1)$ a MIQP, which is outside the scope of this text.

Lemma 2.7 (Convexity tests for Q). *Let $Q = Q^\top$.*

- (a) If $Q \succeq 0$ then $Q_{ii} \geq 0$ for all i , and $Q_{ii} = 0 \Rightarrow Q_{ij} = 0$ for all j .
- (b) (Gershgorin) If $Q_{ii} \geq \sum_{j \neq i} |Q_{ij}|$ for all i , then $Q \succeq 0$.
- (c) $Q \succeq 0$ iff Q has no negative eigenvalue; equivalently, the $LDL^\top$ factorization of $Q + \delta I$ has no negative pivot for every $\delta > 0$ (in exact arithmetic; a computation tests one fixed small δ , which certifies only $Q + \delta I \succeq 0$).

Conditions (a) are necessary, (b) is sufficient; neither is a certificate of convexity in general [7, 8].

3 Duality and Optimality Theory

3.1 Linear programming

Definition 3.1 (Primal–dual pair, bounded form). For $\min\{c^\top x : Ax = b, l \leq x \leq u\}$ the dual is

$$\max_{y, z^l, z^u} b^\top y + l^\top z^l - u^\top z^u \quad \text{s.t.} \quad A^\top y + z^l - z^u = c, \quad z^l, z^u \geq 0,$$

with reduced cost $s = c - A^\top y = z^l - z^u$ (terms with infinite bounds are omitted and the corresponding multiplier is fixed to 0). For the standard form $\{Ax = b, x \geq 0\}$ the dual is $\max\{b^\top y : A^\top y + s = c, s \geq 0\}$.

Theorem 3.2 (Weak duality [1, 2]). For any primal feasible x and dual feasible (y, z^l, z^u) , $c^\top x \geq b^\top y + l^\top z^l - u^\top z^u$.

Theorem 3.3 (Strong duality [1, 2]). If either the primal or the dual has a finite optimal value, both have optimal solutions and the optimal values are equal. Exactly one of the following holds: both have optimal solutions; the primal is unbounded and the dual infeasible; the dual is unbounded and the primal infeasible; both are infeasible.

Theorem 3.4 (Complementary slackness [1, 2]). Feasible x and (y, s) for the standard-form pair are both optimal iff $x_j s_j = 0$ for all j . For the bounded form: $z_j^l(x_j - l_j) = 0$ and $z_j^u(u_j - x_j) = 0$ for all j .

Theorem 3.5 (Goldman–Tucker [12, 1]). If the LP has an optimal solution, there exists an optimal primal–dual pair (x^*, s^*) that is strictly complementary: $x_j^* + s_j^* > 0$ for all j . The optimal face is partitioned into the index sets $\{j : x_j^* > 0\}$ and $\{j : s_j^* > 0\}$, independent of the chosen strictly complementary pair.

Theorem 3.6 (LP optimality (KKT) conditions). x is optimal for $\min\{c^\top x : Ax = b, x \geq 0\}$ iff there exist y, s with

$$Ax = b, \quad A^\top y + s = c, \quad x \geq 0, \quad s \geq 0, \quad x_j s_j = 0 \quad \forall j.$$

3.2 Convex quadratic programming

Theorem 3.7 (QP optimality (KKT) conditions [4, 3]). For convex $Q \succeq 0$, x is optimal for $\min\{\frac{1}{2}x^\top Qx + c^\top x : Ax = b, x \geq 0\}$ iff there exist y, s with

$$Ax = b, \quad A^\top y + s - Qx = c, \quad x \geq 0, \quad s \geq 0, \quad x_j s_j = 0 \quad \forall j.$$

KKT conditions are necessary and sufficient because constraints are linear and the objective is convex.

Definition 3.8 (Wolfe dual of the QP).

$$\max_{x, y, s} b^\top y - \frac{1}{2}x^\top Qx \quad \text{s.t.} \quad A^\top y + s - Qx = c, \quad s \geq 0.$$

Weak and strong duality hold with duality gap $\frac{1}{2}x^\top Qx + c^\top x - (b^\top y - \frac{1}{2}x^\top Qx) = x^\top s$ at a primal–dual feasible point [4, 1].

3.3 Infeasibility and unboundedness

Theorem 3.9 (Farkas lemma [2, 1]). *Exactly one of the following systems has a solution:*

$$(I) \ Ax = b, \ x \geq 0 \quad (II) \ A^\top y \leq 0, \ b^\top y > 0.$$

Equivalently, replacing y by $-y$, (II) reads $A^\top y \geq 0, \ b^\top y < 0$; the two forms are the same statement.

Theorem 3.10 (Variants of the Farkas lemma). *Exactly one of the following holds: (i) $Ax \leq b$ has a solution; (ii) $\exists y \geq 0$ with $A^\top y = 0$ and $b^\top y < 0$. For the bounded system $\{Ax = b, \ l \leq x \leq u\}$, infeasibility is equivalent to the existence of y with $b^\top y > \sum_j \max\{(A^\top y)_j u_j, (A^\top y)_j l_j\}$ (finite bounds assumed).*

Theorem 3.11 (Unboundedness ray). *A feasible LP $\min\{c^\top x : Ax = b, x \geq 0\}$ is unbounded iff there exists $d \geq 0$ with $Ad = 0$ and $c^\top d < 0$. A feasible convex QP is unbounded below iff there exists d in the recession cone of the feasible set with $Qd = 0$ and $c^\top d < 0$ [3].*

Theorem 3.12 (Dual-simplex infeasibility certificate). *If the dual simplex method selects a leaving row r with primal-infeasible basic value and no entering column admits a valid ratio test (row r of $B^{-1}N$ has the correct sign pattern for every nonbasic j at its bound), then $y = \rho_r$, with $\rho_r^\top = e_r^\top B^{-1}$, satisfies $A^\top y \geq 0$ and $b^\top y < 0$. Hence $y = -\rho_r$ satisfies system (II) of the Farkas lemma ($A^\top y \leq 0, \ b^\top y > 0$): the primal LP is infeasible and $-\rho_r$ is a certificate [1, 19].*

3.4 Optimality for MILP

Definition 3.13 (Bound and gap). Let z^{inc} be the objective value of the best integer-feasible point found (upper bound) and $z^{\text{bd}} = \min_{\text{open nodes } k} z_k^{\text{LP}}$ (global lower bound). The relative gap is

$$\text{gap} = \frac{z^{\text{inc}} - z^{\text{bd}}}{\max\{|z^{\text{inc}}|, \varepsilon\}}.$$

An incumbent is proven optimal when the open-node set is empty (gap = 0); otherwise the pair $(z^{\text{inc}}, z^{\text{bd}})$ is a certified bracket of z^* .

Remark 3.14. MILP has no strong duality and no KKT-type optimality system; a residual test can verify only feasibility and integrality, while optimality is certified solely by the bound–incumbent gap.

4 Problem Transformations

4.1 Scaling

Proposition 4.1 (Scaled problem [1]). *Let D_1, D_2 be positive diagonal. Define $\tilde{A} = D_1 A D_2$, $\tilde{b} = D_1 b$, $\tilde{c} = D_2 c$, $\tilde{Q} = D_2 Q D_2$, $\tilde{l} = D_2^{-1} l$, $\tilde{u} = D_2^{-1} u$. Then x is feasible (optimal) for the original problem iff $\tilde{x} = D_2^{-1} x$ is feasible (optimal) for the scaled problem, with identical objective value, and duals map back by*

$$x = D_2 \tilde{x}, \quad y = D_1 \tilde{y}, \quad s = D_2^{-1} \tilde{s}.$$

For QP, symmetric scaling $D_2 Q D_2$ preserves $Q \succeq 0$ (congruence).

Proposition 4.2 (Integer columns). *If $x_j \in \mathbb{Z}$ and column j is scaled by $d_j > 0$, then $\tilde{x}_j = x_j/d_j \in d_j^{-1}\mathbb{Z}$; $\tilde{x}_j$ is integer for all integer x_j only if $d_j^{-1} \in \mathbb{Z}$. The integer-feasible sets of the original and scaled problems correspond exactly iff $d_j = 1$: if $d_j^{-1} \notin \mathbb{Z}$, integrality of x_j is not preserved; if $d_j^{-1} \in \mathbb{Z}$ with $d_j^{-1} > 1$, the scaled problem admits integer $\tilde{x}_j$ corresponding to fractional x_j , enlarging the integer set. Hence integer columns must be excluded from column scaling. Row scaling does not affect integrality.*

Method 4.3 (Ruiz equilibration [20]). Set $A^{(0)} = A$. For $k = 0, 1, \dots$: $r_i = \sqrt{\|\text{row}_i A^{(k)}\|_\infty}$, $c_j = \sqrt{\|\text{col}_j A^{(k)}\|_\infty}$, $A^{(k+1)} = \text{diag}(r)^{-1} A^{(k)} \text{diag}(c)^{-1}$. Then $D_1 = \prod_k \text{diag}(r^{(k)})^{-1}$ and $D_2 = \prod_k \text{diag}(c^{(k)})^{-1}$, and all row and column ∞ -norms of $A^{(k)}$ converge to 1.

Method 4.4 (Pock–Chambolle preconditioning [21]). For $\alpha \in [0, 2]$ define step-size matrices $T = \text{diag}(\tau_j)$, $\Sigma = \text{diag}(\sigma_i)$ with

$$\tau_j = \left(\sum_i |a_{ij}|^{2-\alpha} \right)^{-1}, \quad \sigma_i = \left(\sum_j |a_{ij}|^\alpha \right)^{-1}.$$

Then $\|\Sigma^{1/2} A T^{1/2}\|_2 \leq 1$, the step-size condition of preconditioned PDHG; convergence in the sense of the PDHG theorem below is guaranteed when the steps are additionally scaled by a factor below one (strict inequality).

4.2 Presolve

Let the row activity bounds of row i be

$$L_i = \sum_j \min\{a_{ij}l_j, a_{ij}u_j\}, \quad U_i = \sum_j \max\{a_{ij}l_j, a_{ij}u_j\}.$$

Proposition 4.5 (Presolve reductions [22, 23]). *For LP and MILP ($Q = 0$), each of the following preserves the optimal value and lets an optimal primal and dual solution of the original problem be recovered (see postsolve). Rows are written $a_i^\top x \leq b_i$; other senses are analogous.*

- R1.** Empty row/column. *An empty row with $0 \leq b_i$ is dropped (infeasible otherwise). An empty column is fixed at the bound that minimizes $c_j x_j$ (unbounded or infeasible if that bound is infinite and $c_j \neq 0$).*
- R2.** Singleton row. *A row $a_{ij}x_j \leq b_i$ is replaced by a bound on x_j : $x_j \leq b_i/a_{ij}$ if $a_{ij} > 0$, $x_j \geq b_i/a_{ij}$ if $a_{ij} < 0$.*
- R3.** Fixed column. *If $l_j = u_j$, substitute $x_j = l_j$ and update $b \leftarrow b - a_j l_j$.*
- R4.** Redundant row. *If $U_i \leq b_i$, drop the row. If $L_i > b_i$, the problem is infeasible.*
- R5.** Forcing row. *If $L_i = b_i$, every x_j with $a_{ij} \neq 0$ must sit at the bound attaining L_i ; fix them and drop the row.*
- R6.** Doubleton equation. *For $a_{i1}x_1 + a_{i2}x_2 = b_i$ substitute $x_1 = (b_i - a_{i2}x_2)/a_{i1}$ everywhere and transfer the bounds of x_1 to x_2 .*
- R7.** Dual fixing. *If $c_j \geq 0$ and every row containing j is an inequality row in which decreasing x_j cannot cause a violation ($a_{ij} \geq 0$ in $\leq$ -rows, $a_{ij} \leq 0$ in $\geq$ -rows; an equality row blocks the argument), then some optimal solution has $x_j = l_j$ when l_j is finite; if $l_j = -\infty$ and $c_j > 0$, the LP is unbounded or infeasible; if $l_j = -\infty$ and $c_j = 0$, the column and every row containing it can be dropped, with x_j chosen in postsolve to satisfy those rows. Symmetrically for $c_j \leq 0$ with upper bounds.*
- R8.** Implied-free column substitution / aggregator. *If column j appears in an equality row i and the bounds of x_j are implied by the bounds of the other variables of row i , then x_j can be eliminated: $x_j = (b_i - \sum_{k \neq j} a_{ik}x_k)/a_{ij}$, with its cost folded into the remaining variables.*

Remark 4.6 (Presolve with $Q \neq 0$). For QP the reductions change as follows. R1: an empty column with a nonzero in Q is not fixed by its cost alone; it is removed only if $Q_{\cdot j} = 0$. R3: substituting $x_j = l_j$ also updates $c \leftarrow c + Q_{\cdot j} l_j$ on the remaining variables and adds the constant $c_j l_j + \frac{1}{2} Q_{jj} l_j^2$ to the objective. R6 and R8: substituting $x_1 = \alpha + \beta x_2$ replaces Q by its congruent transform, with corresponding changes to c and the constant. R7: dual fixing is invalid when column j has a nonzero in Q . Reduced costs are recovered as $s = c + Qx - A^\top y$.

Proposition 4.7 (Bound tightening from rows [23]). *For $a_{ik} > 0$ in $a_i^\top x \leq b_i$: $x_k \leq (b_i - L_i^{(-k)})/a_{ik}$, where $L_i^{(-k)} = L_i - \min\{a_{ik}l_k, a_{ik}u_k\}$. For $a_{ik} < 0$ the same expression gives a lower bound on x_k . Tightened bounds are valid for all feasible points.*

Proposition 4.8 (MILP-specific reductions [23, 46]). (a) Integer bound rounding: for $j \in \mathcal{I}$, $l_j \leftarrow \lceil l_j \rceil$, $u_j \leftarrow \lfloor u_j \rfloor$.

(b) Coefficient (GCD) tightening: if all variables of $\sum_j a_j x_j \leq b$ are integer with integer coefficients and $g = \gcd(a_j)$, then the row may be replaced by $\sum_j (a_j/g) x_j \leq \lfloor b/g \rfloor$. For an equality, the row is infeasible unless $g \mid b$.

(c) Reduced integer columns: an integer column must not be eliminated by substitution (R6, R8) unless the substituted expression is integer-valued for all integer inputs.

Definition 4.9 (Postsolve). Postsolve applies the inverse of each reduction in reverse order (last in, first out), recovering: primal values of removed variables (from the recorded substitution or fixed bound); row duals y_i of removed rows (zero for redundant rows; for a singleton row whose implied bound is active at x_j (no column is eliminated), set to make the reduced cost s_j zero; for a doubleton equation, set to make the reduced cost of the eliminated column zero); and reduced costs $s_j = c_j - a_j^\top y$ for removed columns, with $s_j \geq 0$ at a lower bound and $s_j \leq 0$ at an upper bound [22].

5 Simplex Theory

Definition 5.1 (Basic solution). For $\{Ax = b, x \geq 0\}$ with $\text{rank } A = m$, a basis $\mathcal{B}_s$ is a set of m indices with $B = A_{\mathcal{B}_s}$ nonsingular. The basic solution is $x_{\mathcal{B}_s} = B^{-1}b$, $x_{\mathcal{N}} = 0$; it is a *basic feasible solution* (BFS) if $x_{\mathcal{B}_s} \geq 0$. The reduced costs are $d_{\mathcal{N}} = c_{\mathcal{N}} - N^\top B^{-1\top} c_{\mathcal{B}_s}$ and the dual estimate is $y = B^{-1\top} c_{\mathcal{B}_s}$.

Theorem 5.2 (Fundamental theorem of LP [2]). If a standard-form LP is feasible it has a BFS; if it has an optimal solution it has an optimal BFS. BFS correspond exactly to vertices (extreme points) of the feasible polyhedron.

Theorem 5.3 (Simplex optimality [1]). If $d_{\mathcal{N}} \geq 0$ at a BFS, the BFS is optimal; the converse holds for a nondegenerate BFS, while a degenerate optimal BFS may have some $d_j < 0$. A basis is dual feasible iff $d_{\mathcal{N}} \geq 0$; it is optimal iff additionally $x_{\mathcal{B}_s} \geq 0$.

Method 5.4 (Primal simplex iteration). Choose entering $q \in \mathcal{N}$ with $d_q < 0$; compute $\alpha = B^{-1}a_q$; ratio test $\theta = \min_{i: \alpha_i > 0} x_{B(i)}/\alpha_i$ picks the leaving row r ; update $x_{\mathcal{B}_s} \leftarrow x_{\mathcal{B}_s} - \theta\alpha$, $x_q = \theta$ and the basis. If $\alpha \leq 0$ the LP is unbounded along the ray $(-\alpha, e_q)$.

Method 5.5 (Dual simplex iteration [1, 19]). Choose leaving row r with $x_{B(r)} < 0$ (primal infeasible). Let $\rho_r^\top = e_r^\top B^{-1}$ and $\alpha_{rj} = \rho_r^\top a_j$ for $j \in \mathcal{N}$. Ratio test over $\{j : \alpha_{rj} < 0\}$ selects entering $q = \arg \min_j d_j/(-\alpha_{rj})$, keeping $d \geq 0$. If no such j exists the primal is infeasible (Theorem 3.12). The dual objective is nondecreasing.

Definition 5.6 (Bounded variables). For variables with bounds $[l_j, u_j]$, a nonbasic variable sits at l_j (dual feasible iff $d_j \geq 0$) or at u_j (iff $d_j \leq 0$). A boxed variable with the wrong-sign reduced cost can be moved to its opposite bound without loss of dual feasibility (*bound flipping*). The iteration formulas of this section are written for nonbasic variables at a lower bound (or $x \geq 0$); for a variable at an upper bound the sign conditions on d_j and α_{rj} are reversed.

Theorem 5.7 (Termination and degeneracy [1, 13]). Each nondegenerate simplex iteration strictly improves the objective, so a method visiting only nondegenerate bases terminates finitely. A pivot is degenerate if the step length is $\theta = 0$; cycling among degenerate bases is possible. **Bland's rule** (choose the smallest-index eligible entering variable and, among ties in the ratio test, the smallest-index leaving variable) guarantees finite termination [13].

Method 5.8 (Harris two-pass ratio test [14]). With feasibility tolerance $\delta > 0$: pass 1 computes $\theta_{\max} = \min_j (x_j + \delta)/\alpha_j$ over $\alpha_j > 0$; pass 2 selects, among $\{j : x_j/\alpha_j \leq \theta_{\max}\}$, the index with the largest $|\alpha_j|$. This bounds the size of the pivot element away from zero at the cost of small bounded infeasibilities of size $\leq \delta$.

Method 5.9 (Bound-flipping (long-step) dual ratio test [19]). Sort the breakpoints $t_j = d_j/|\alpha_{rj}|$. Traverse them in order; at each breakpoint of a boxed variable j , the slope of the dual objective decreases by $|\alpha_{rj}|(u_j - l_j)$; continue while the slope remains positive. All passed boxed variables are flipped to their opposite bounds and the entering variable is chosen from the final group (largest $|\alpha|$).

Proposition 5.10 (Cost perturbation [19]). *Let $\tilde{c} = c + \varepsilon$ with ε_j small. The feasible set is unchanged, so any basis primal feasible for the perturbed problem is primal feasible for the original. If a basis is optimal for the perturbed problem, then removing the perturbation leaves it primal feasible, and primal simplex iterations from it (cleanup) reach an optimum of the original problem. Perturbation makes dual degeneracy (ties in ratio tests $d_j = 0$) unlikely. (Bound perturbation, the EXPAND variant, is treated in [18].)*

Method 5.11 (Dual phase 1 by artificial bounding [19]). Solve the auxiliary problem with the original costs, $b = 0$, and replaced bounds: free $\rightarrow [-1, 1]$, lower-bounded only $\rightarrow [0, 1]$, upper-bounded only $\rightarrow [-1, 0]$, boxed or fixed $\rightarrow [0, 0]$. Every basis is dual feasible for it after bound flips, and its optimal value equals minus the minimum, over all dual vectors y , of the sum of dual infeasibilities of the original problem; this minimum is attained at the final basis of the auxiliary problem, not at the starting basis. An optimal value of 0 yields a dual feasible basis for the original problem; otherwise the original LP is dual infeasible [19].

Method 5.12 (Primal phase 1 [1]). Minimize the sum of infeasibilities $w = \sum_i \max\{0, l_i - x_{B(i)}, x_{B(i)} - u_i\}$; the piecewise-linear objective is handled by a breakpoint ratio test. The LP is feasible iff $\min w = 0$.

5.1 Basis factorization and pricing

Definition 5.13 (LU with Markowitz pivoting [1, 17]). A row/column-permuted factorization $PBQ = LU$ with L unit lower triangular. At each elimination step the pivot b_{ij} minimizes the Markowitz count $(r_i - 1)(c_j - 1)$ (row and column counts in the active submatrix) among entries satisfying the threshold test $|b_{ij}| \geq u_{\text{piv}} \max_k |b_{kj}|$, with $0 < u_{\text{piv}} \leq 1$.

Theorem 5.14 (Forrest–Tomlin update [16]). *If B' differs from B in one column, then $L^{-1}B'$ is upper triangular except in one spike column. A row-eta transformation removes the spike, giving $B' = L R_k^{-1} U'$ with R_k an elementary row-eta matrix and U' upper triangular, at a cost proportional to the nonzeros of the spike. After a bounded number of updates the factorization is recomputed.*

Theorem 5.15 (Dual steepest-edge weights [15]). *Define $\beta_i = \|e_i^\top B^{-1}\|_2^2$. With pivot row r , entering column $\alpha = B^{-1}a_q$ and $\tau = B^{-1}\rho_r$, the updated weights are*

$$\beta'_r = \frac{\beta_r}{\alpha_r^2}, \quad \beta'_i = \beta_i - 2\frac{\alpha_i}{\alpha_r}\tau_i + \left(\frac{\alpha_i}{\alpha_r}\right)^2 \beta_r \quad (i \neq r),$$

and the leaving row maximizes $(x_{B(i)}^{\text{infeas}})^2/\beta_i$. Primal DEX uses approximate reference weights instead [15, 14].

6 First-Order Method: PDHG and PDLP

Definition 6.1 (Saddle-point form). For $\min\{c^\top x : Ax \{=, \geq\} b, l \leq x \leq u\}$ let $X = \{l \leq x \leq u\}$ and $Y = \{y : y_i \geq 0 \text{ for inequality rows}\}$. Then

$$\min_{x \in X} \max_{y \in Y} \mathcal{L}(x, y) = c^\top x + y^\top (b - Ax).$$

Method 6.2 (PDHG [24, 25]). With primal weight $\omega > 0$, step size $\eta > 0$, $\tau = \eta/\omega$, $\sigma = \eta\omega$:

$$x^{k+1} = \text{proj}_X(x^k - \tau(c - A^\top y^k)), \quad y^{k+1} = \text{proj}_Y(y^k + \sigma(b - A(2x^{k+1} - x^k))).$$

Theorem 6.3 (Convergence of PDHG [24]). *If $\tau\sigma\|A\|_2^2 < 1$, the iterates converge to a saddle point and the ergodic averages satisfy an $O(1/k)$ rate for the primal–dual gap.*

Theorem 6.4 (Restarted PDHG for LP [26]). *The sequence of restarted ergodic averages of PDHG (restart based on the normalized duality gap) converges linearly for LP, with a rate depending on a sharpness constant related to the Hoffman constant of the KKT system, in contrast to the sublinear rate of unrestarted PDHG. This is proved for restarts based on the normalized duality gap; the KKT-error restarts of the practical variant below are not strictly covered.*

Method 6.5 (PDLP enhancements [25, 26]). (a) *Adaptive step size.* Accept a trial step η iff

$$\eta \leq \bar{\eta} = \frac{\omega\|\Delta x\|^2 + \|\Delta y\|^2/\omega}{2|\Delta y^\top A \Delta x|}, \text{ then set } \eta' = \min\{(1 - (k+1)^{-0.3})\bar{\eta}, (1 + (k+1)^{-0.6})\eta\}.$$

(b) *Step-size-weighted ergodic average* of the iterates.

(c) *Adaptive restarts* to the better (in KKT error) of the current iterate and the average when: sufficient decay $\text{KKT}(\text{cand}) \leq \beta_{\text{suff}} \text{KKT}(\text{last restart})$; or necessary decay together with no further progress (β_{nec}); or after an artificial fraction β_{art} of the total iterations. The published values are $\beta_{\text{suff}} = 0.2$, $\beta_{\text{nec}} = 0.8$, $\beta_{\text{art}} = 0.36$.

(d) *Primal weight update* at each restart: $\omega \leftarrow \exp(\theta \log(\|\Delta y\|/\|\Delta x\|) + (1-\theta) \log \omega)$, $\theta = 0.5$.

(e) *Presolve and diagonal preconditioning* (Ruiz followed by Pock–Chambolle).

Definition 6.6 (KKT error and termination). For an iterate (x, y) with reduced costs s , the relative primal residual, dual residual and gap

$$\frac{\|\text{viol}(Ax - b)\|}{1 + \|b\|}, \quad \frac{\|\text{viol}(c - A^\top y - s)\|}{1 + \|c\|}, \quad \frac{|c^\top x - \text{dual objective}|}{1 + |c^\top x| + |\text{dual objective}|}$$

are all $\leq \epsilon$ at termination; this is evaluated on the *unscaled* problem. Norms here are Euclidean, as in PDLP.

Remark 6.7. PDHG returns an approximately optimal point (accuracy $\epsilon \approx 10^{-4}$), generally not a vertex; a vertex requires crossover (Section 7).

7 Crossover

Theorem 7.1 (Existence of a vertex at least as good [27]). *For any optimal primal solution $\bar{x}$ of an LP there exists a basic optimal solution x^B in which every variable at a bound in $\bar{x}$ is at the same bound (for $x \geq 0$ this is $\text{supp}(x^B) \subseteq \text{supp}(\bar{x})$). Given an optimal primal–dual pair, a basic optimal pair can be computed in strongly polynomial time by a sequence of at most n primal pushes and m dual pushes.*

Method 7.2 (Primal push [27, 28]). Let $\bar{x}$ be feasible with nonbasic-labelled variables at bounds. If a variable strictly between bounds is still labelled nonbasic, find a direction d with $Ad = 0$, $d_j = 1$ (supported on basic and the pushed variable), and step until a variable reaches a bound; that variable becomes nonbasic, the pushed variable enters the basis. The objective does not increase because $c^\top d = 0$ at optimality (complementary slackness).

Method 7.3 (Basis identification [28]). From a near-optimal $(\bar{x}, \bar{y}, \bar{s})$, classify variables using complementarity: $\bar{x}_j - l_j > \bar{s}_j$ suggests basic, otherwise nonbasic at a bound. A nonsingular basis is then obtained by a crash procedure that inserts structural columns in order of “basicness” replacing logical (slack) columns only, keeping the basis nonsingular at every step. Dual simplex or primal simplex then finishes from this basis.

8 Interior-Point Methods for LP and QP

Definition 8.1 (Perturbed KKT and central path [1, 11]). For the standard-form QP (LP when $Q = 0$) and $\mu > 0$:

$$Ax = b, \quad A^\top y + s - Qx = c, \quad XSe = \mu e, \quad x > 0, \quad s > 0.$$

The solutions (x_μ, y_μ, s_μ) , $\mu > 0$, form the *central path*. The duality measure is $\mu = x^\top s/n$.

Theorem 8.2 (Central path [11, 1]). *If the strictly feasible set $\{Ax = b, A^\top y + s - Qx = c, x > 0, s > 0\}$ is nonempty, then for every $\mu > 0$ the central-path system has a unique solution (x_μ, s_μ) (unique y_μ when A has full row rank). As $\mu \rightarrow 0$ the path converges to a strictly complementary optimal solution (for LP, the analytic center of the optimal face; for QP only strict complementarity is asserted).*

Method 8.3 (Newton step). With residuals $r_p = b - Ax$, $r_d = c - A^\top y - s + Qx$, $r_c = \mu e - XSe$, the Newton direction solves

$$A\Delta x = r_p, \quad -Q\Delta x + A^\top \Delta y + \Delta s = r_d, \quad S\Delta x + X\Delta s = r_c.$$

Eliminating $\Delta s = X^{-1}(r_c - S\Delta x)$ gives the *augmented system*

$$\begin{bmatrix} -(Q + X^{-1}S) & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} r_d - X^{-1}r_c \\ r_p \end{bmatrix}.$$

For LP, further elimination gives the normal equations $A(X^{-1}S)^{-1}A^\top \Delta y = \dots$.

Definition 8.4 (Inertia and quasidefinite matrices [31]). The inertia $\text{In}(K) = (n_+, n_-, n_0)$ counts positive, negative and zero eigenvalues of symmetric K . A symmetric matrix $\begin{bmatrix} -H & A^\top \\ A & G \end{bmatrix}$ with $H \succ 0$ and $G \succ 0$ is *quasidefinite*.

Theorem 8.5 (Factorizability of quasidefinite matrices [31]). *Every symmetric permutation $PKP^\top$ of a quasidefinite K has an $LDL^\top$ factorization with diagonal D (no 2×2 pivots) and n negative and m positive pivots, with stability depending on the regularization size. Hence the fill-reducing ordering can be chosen freely.*

Theorem 8.6 (Inertia of the KKT matrix [4, 34]). *Let $K = \begin{bmatrix} -H & A^\top \\ A & 0 \end{bmatrix}$ with $H \in \mathbb{R}^{n \times n}$ symmetric and A of full row rank. Then $\text{In}(K) = (m, n, 0)$ iff H is positive definite on $\ker A$. For the IPM matrix $H = Q + X^{-1}S$ with $Q \succeq 0$ and $x, s > 0$, $H \succ 0$, and so the inertia is $(m, n, 0)$ whenever A has full row rank. A different inertia certifies rank deficiency or negative curvature of H on $\ker A$; the correct inertia does not certify $Q \succeq 0$, since $X^{-1}S \succ 0$ can mask negative curvature of Q .*

Method 8.7 (Regularization [32, 33]). Replace K by $K_\delta = \begin{bmatrix} -(H + \delta_p I) & A^\top \\ A & \delta_d I \end{bmatrix}$, $\delta_p, \delta_d > 0$.

This is quasidefinite; it corresponds to one step of a proximal-point method on primal and dual variables. If a pivot has the wrong sign during $LDL^\top$, replace it by $\text{sign} \cdot \max(|\text{pivot}|, \delta)$ (*dynamic regularization*). The bias introduced is removed by iterative refinement against the unregularized K (Section 11).

Method 8.8 (Mehrotra predictor-corrector [29]). (1) Solve the Newton system with $r_c = -XSe$ (affine step Δ_{aff}). (2) Compute α_{aff} by fraction-to-boundary, $\mu_{\text{aff}} = (x + \alpha_p \Delta x_{\text{aff}})^\top (s + \alpha_d \Delta s_{\text{aff}})/n$ and centering parameter $\sigma = (\mu_{\text{aff}}/\mu)^3$. (3) Solve with

$$r_c = \sigma \mu e - XSe - \Delta X_{\text{aff}} \Delta S_{\text{aff}} e,$$

reusing the same factorization.

Method 8.9 (Gondzio multiple centrality correctors [30]). After the predictor–corrector direction Δ , compute trial step $\tilde{\alpha} = \min(1, \alpha + \delta_\alpha)$ and trial complementarity products $v = (X + \tilde{\alpha}\Delta X)(S + \tilde{\alpha}\Delta S)e$. Project v onto the box $[\beta_{\min}\mu, \beta_{\max}\mu]$ to obtain a target t , and solve the Newton system with $r_c \leftarrow t - v$ (same factorization). Accept the corrected direction if the step length increases sufficiently; repeat up to K times.

Method 8.10 (Step-length rules [4, 11]). The fraction-to-the-boundary rule: $\alpha_p = \min(1, \eta\alpha_p^{\max})$, $\alpha_d = \min(1, \eta\alpha_d^{\max})$ with $\eta \in (0, 1)$, where $\alpha^{\max}$ is the largest step keeping $x, s \geq 0$. For LP separate primal and dual steps are valid. For QP the dual residual $Qx + c - A^\top y - s$ depends on both, so $\alpha_p = \alpha_d$ is used, which keeps the linear residuals decreasing proportionally by $(1 - \alpha)$.

Proposition 8.11 (Residual contraction). *After a step α along a Newton direction obtained by an exact solve, the linear residuals satisfy $r_p^+ = (1 - \alpha)r_p$ and $r_d^+ = (1 - \alpha)r_d$ (for QP with $\alpha_p = \alpha_d = \alpha$); with regularization or inexact solves this holds only up to the linear-solve error.*

Theorem 8.12 (Complexity of path-following [11]). *Feasible-start path-following methods with neighborhood-preserving steps reach $x^\top s \leq \epsilon$ in $O(\sqrt{n} \log(1/\epsilon))$ iterations (short-step) or $O(n \log(1/\epsilon))$ (long-step). Infeasible-start variants have weaker bounds, roughly $O(n \log(1/\epsilon))$ (short-step) and $O(n^2 \log(1/\epsilon))$ (long-step), where ϵ bounds both the residual norms and $x^\top s$. Mehrotra’s method is used in practice without a polynomial guarantee.*

Remark 8.13 (Sparse LDL[⊤] and ordering [10, 35]). Fill-in of a sparse Cholesky/LDL[⊤] factorization depends on the elimination order. Finding the minimum-fill ordering is NP-hard; approximate minimum degree (AMD) is the standard heuristic. Existence of the quasidefinite LDL[⊤] factorization under arbitrary ordering is given by Theorem 8.5; its numerical stability depends on the size of the regularization.

Lemma 8.14 (Dependent equality rows). *If the equality rows of A are linearly dependent, the system $Ax = b$ is either inconsistent or contains redundant rows; redundant rows can be removed without changing the feasible set, and dependence makes K singular. Removing linearly dependent rows (found by row-echelon reduction) restores full row rank.*

9 ADMM for Quadratic Programming

Definition 9.1 (Splitting [37]). The QP $\min\{\frac{1}{2}x^\top Qx + c^\top x : l \leq Ax \leq u\}$ is rewritten as

$$\min_{x,z} \frac{1}{2}x^\top Qx + c^\top x + \mathbb{I}_{[l,u]}(z) \quad \text{s.t.} \quad Ax = z.$$

Method 9.2 (ADMM iteration [37, 36]). With parameters $\sigma, \rho > 0$:

$$(Q + \sigma I + \rho A^\top A)x^{k+1} = \sigma x^k - c + A^\top(\rho z^k - y^k), \quad z^{k+1} = \text{proj}_{[l,u]}(Ax^{k+1} + y^k / \rho), \quad y^{k+1} = y^k + \rho(Ax^{k+1} - z^{k+1}).$$

The linear system may be solved by preconditioned conjugate gradients using only products with Q , A , $A^\top$.

Theorem 9.3 (Convergence [36, 37]). *For convex $Q \succeq 0$ and $\rho, \sigma > 0$ the ADMM iterates converge to a KKT point when the QP is solvable; otherwise the differences $\delta x = \lim(x^{k+1} - x^k)$, $\delta y = \lim(y^{k+1} - y^k)$ exist.*

Theorem 9.4 (Infeasibility and unboundedness detection [38]). *Let δy and δx be the limits above.*

- (a) *If $\delta y \neq 0$ satisfies $A^\top \delta y = 0$ and $u^\top \max(\delta y, 0) + l^\top \min(\delta y, 0) < 0$, these conditions certify primal infeasibility; conversely, for an infeasible QP the limit δy satisfies them.*
- (b) *If $\delta x \neq 0$, the QP is dual infeasible iff $Q\delta x = 0$, $c^\top \delta x < 0$ and $A\delta x$ lies in the recession cone of $[l, u]$; if the QP is also feasible, it is unbounded below.*

10 Mixed-Integer Linear Programming Theory

Theorem 10.1 (Relaxation bound [5]). *For the MILP $z^* = \min\{c^\top x : x \in P, x_j \in \mathbb{Z} (j \in \mathcal{I})\}$ with LP relaxation value $z^{\text{LP}} = \min\{c^\top x : x \in P\}$, $z^{\text{LP}} \leq z^*$. If the LP optimum is integer feasible it is optimal for the MILP.*

10.1 Cutting planes

Definition 10.2 (Valid inequality, cut). An inequality $\pi^\top x \geq \pi_0$ is *valid* if it holds for all integer-feasible points; it is a *cut* at LP optimum x^* if additionally $\pi^\top x^* < \pi_0$.

Theorem 10.3 (Gomory mixed-integer cut [6, 39]). *Let a simplex tableau row with nonnegative nonbasic variables read $x_i + \sum_{j \in \mathcal{N}} \bar{a}_j x_j = \bar{b}$ with $x_i \in \mathbb{Z}$ and $f_0 = \bar{b} - \lfloor \bar{b} \rfloor \in (0, 1)$. With $f_j = \bar{a}_j - \lfloor \bar{a}_j \rfloor$, the inequality*

$$\sum_{\substack{j \in \mathcal{I} \cap \mathcal{N} \\ f_j \leq f_0}} f_j x_j + \sum_{\substack{j \in \mathcal{I} \cap \mathcal{N} \\ f_j > f_0}} \frac{f_0(1-f_j)}{1-f_0} x_j + \sum_{\substack{j \notin \mathcal{I} \\ \bar{a}_j > 0}} \bar{a}_j x_j + \sum_{\substack{j \notin \mathcal{I} \\ \bar{a}_j < 0}} \frac{f_0}{1-f_0} (-\bar{a}_j) x_j \geq f_0$$

is valid for the MILP and cuts off the LP vertex.

Theorem 10.4 (Mixed-integer rounding [40, 6]). *Let $X = \{(x, s) \in \mathbb{Z}_+^n \times \mathbb{R}_+ : \sum_j a_j x_j - s \leq b\}$, $f = b - \lfloor b \rfloor > 0$, $f_j = a_j - \lfloor a_j \rfloor$. Then*

$$\sum_j \left(\lfloor a_j \rfloor + \frac{(f_j - f)^+}{1-f} \right) x_j - \frac{1}{1-f} s \leq \lfloor b \rfloor$$

is valid for X . Complemented MIR (c-MIR) [41] applies it after (i) aggregating rows $\sum_i \lambda_i (a_i^\top x \leq b_i)$ (multi-row aggregation), (ii) substituting variable bounds $x_j = l_j + x'_j$ or $x_j = u_j - x'_j$ (complementing), (iii) substituting variable-upper-bound relations $x_j \leq u_j y_j$ for continuous x_j , and (iv) scaling the row by $\delta > 0$; every choice yields a valid inequality.

Theorem 10.5 (Cover inequalities [42, 43]). *For a knapsack row $\sum_j a_j x_j \leq b$, $a_j > 0$, $x \in \{0, 1\}^n$, a cover C satisfies $\sum_{j \in C} a_j > b$, and $\sum_{j \in C} x_j \leq |C| - 1$ is valid. It is strengthened by lifting: $\sum_{j \in C} x_j + \sum_{j \notin C} \alpha_j x_j \leq |C| - 1$ with sequentially computed $\alpha_j \geq 0$ (extended cover when $\alpha_j = 1$ for $a_j \geq \max_{k \in C} a_k$).*

Theorem 10.6 (Clique inequalities [45]). *For a conflict graph on binaries and their complements (edge = pair that cannot both equal 1), every clique K gives the valid inequality $\sum_{j \in K} \tilde{x}_j \leq 1$, where $\tilde{x}_j \in \{x_j, 1 - x_j\}$.*

Theorem 10.7 (Flow cover inequalities [44]). *For the single-node flow set $\{(x, y) : \sum_{j \in N_1} x_j - \sum_{j \in N_2} x_j \leq b, 0 \leq x_j \leq u_j y_j, y_j \in \{0, 1\}\}$ and a flow cover (C_1, C_2) with $\lambda = \sum_{C_1} u_j - \sum_{C_2} u_j - b > 0$, a family of lifted flow cover inequalities in (x, y) is valid. Flow covers can be separated by substituting variable upper bounds and applying MIR (path/flow-cover strength [41]).*

Lemma 10.8 (Cut violation). *For a cut $\pi^\top x \geq \pi_0$ and LP point x^* , the (Euclidean-normalized) violation $(\pi_0 - \pi^\top x^*) / \|\pi\|_2$ is the distance of x^* to the cut hyperplane; cuts with large violation and large mutual angle (near-orthogonal cuts) are preferred, and near-parallel cuts are redundant.*

10.2 Probing and propagation

Proposition 10.9 (Bound propagation). *If tightened bounds are derived from row activity bounds (Proposition 4.7), then every feasible point of the original problem satisfies the tightened bounds. Iteration reaches a fixed point in a finite number of passes when applied to integer variables with finite bounds.*

Theorem 10.10 (Probing [46]). For a binary x_j , let P_v be the result of fixing $x_j = v$ and propagating. (i) If P_0 proves infeasibility, then $x_j = 1$ in every feasible solution (and vice versa). (ii) If both P_0 and P_1 tighten x_k , then $x_k \leq \max(u_k^{(0)}, u_k^{(1)})$ and $x_k \geq \min(l_k^{(0)}, l_k^{(1)})$ hold globally. (iii) If P_v fixes $x_k = w$, then $x_j = v \Rightarrow x_k = w$ is a valid implication.

10.3 Branching and tree search

Proposition 10.11 (Branching disjunction). If $x_j^* \notin \mathbb{Z}$ for $j \in \mathcal{I}$, then $\{x_j \leq \lfloor x_j^* \rfloor\} \cup \{x_j \geq \lceil x_j^* \rceil\}$ contains all integer-feasible points; the two children partition the feasible integer set. When integer variables are bounded, the tree is finite.

Definition 10.12 (Pseudocost and score [47]). Let $\Delta_j^\pm$ be observed objective increases when branching down/up on j , per unit of fractionality change. The pseudocosts Ψ_j^-, Ψ_j^+ are the averages of Δ_j^-/f_j and $\Delta_j^+/(1 - f_j)$ with $f_j = x_j^* - \lfloor x_j^* \rfloor$. The estimated degradations are $\hat{\Delta}_j^- = \Psi_j^- f_j$, $\hat{\Delta}_j^+ = \Psi_j^+ (1 - f_j)$, and the product score

$$\text{score}_j = \max\{\hat{\Delta}_j^-, \varepsilon\} \cdot \max\{\hat{\Delta}_j^+, \varepsilon\}$$

is maximized. *Strong branching* replaces $\hat{\Delta}_j^\pm$ by the objective increases of truncated child LPs. *Reliability branching* [47] uses strong branching until $\min(\eta_j^-, \eta_j^+) \geq \eta_{\text{rel}}$ (number of observations), then pseudocosts.

Theorem 10.13 (Pruning rules [5]). A node may be discarded if: (i) its LP is infeasible; (ii) its LP value $z_k \geq z^{\text{inc}} - \text{tol}$ (bound no better than the incumbent); or (iii) its LP solution is integer feasible (the incumbent is updated if better). Rules (i) and (iii) never remove a better integer-feasible solution, and (ii) removes none when the tolerance is 0; with a positive tolerance, solutions better than the incumbent by less than the tolerance may be discarded.

Theorem 10.14 (Reduced-cost fixing [5]). Let the node LP have optimal value z^{LP} , and let nonbasic x_j at its lower bound have reduced cost $d_j > 0$. Then every integer-feasible point with objective $< z^{\text{inc}}$ satisfies

$$x_j \leq l_j + \left\lfloor \frac{z^{\text{inc}} - z^{\text{LP}}}{d_j} \right\rfloor \quad (j \in \mathcal{I}),$$

and symmetrically $x_j \geq u_j - \lfloor (z^{\text{inc}} - z^{\text{LP}})/|d_j| \rfloor$ for $d_j < 0$ at an upper bound. These bounds are valid in the entire subtree.

10.4 Primal heuristics

Definition 10.15 (Heuristic guarantee). Primal heuristics output points that are checked for feasibility and integrality; an accepted point gives a valid *upper bound* z^{inc} . No optimality guarantee is provided by a heuristic; the lower bound comes only from relaxations.

Method 10.16 (Feasibility pump [48]). Alternate: (1) round the LP point x^* to $\tilde{x} = \lfloor x^* \rfloor$ on integer coordinates; (2) solve the LP $\min \Delta(x, \tilde{x}) = \sum_{j \in \mathcal{I}_{\text{bin}}} \tilde{x}_j x_j + \sum_{j \in \mathcal{I}_{\text{bin}}} \tilde{x}_j (1 - x_j)$ (generalized to general integers with auxiliary variables) over the relaxation. Stop when $\Delta = 0$; perturb the rounding upon detecting cycles.

Method 10.17 (RINS and RENS [49, 50]). *RINS*: given incumbent $\bar{x}$ and node LP point x^* , fix all integer variables with $\bar{x}_j = x_j^*$ and solve the resulting sub-MILP. *RENS*: restrict each integer variable to $[\lfloor x_j^* \rfloor, \lceil x_j^* \rceil]$ (fixing those already integral); no incumbent is required.

Method 10.18 (Diving / rounding / fix-and-propagate). Diving repeatedly fixes (or bounds) the most promising fractional variable, propagates bounds, and re-solves the LP until integer feasible or infeasible. Simple rounding accepts an LP point rounded coordinatewise if the round direction cannot violate any row (*locks*); fix-and-propagate fixes integer variables one at a time, propagating after each fixing [45, 50].

11 Numerical Analysis

Definition 11.1 (Condition number). For nonsingular M , $\kappa(M) = \|M\|\|M^{-1}\|$. A perturbation δb of the right-hand side of $Mx = b$ satisfies $\|\delta x\|/\|x\| \leq \kappa(M)\|\delta b\|/\|b\|$.

Theorem 11.2 (Backward stability of LU [9]). *Gaussian elimination with partial (or threshold) pivoting produces computed $\hat{x}$ with $(M + \Delta M)\hat{x} = b$, $\|\Delta M\| \leq c_n \rho_n \epsilon_{\text{mach}}\|M\|$, where ρ_n is the growth factor. Threshold pivoting bounds ρ_n through the threshold parameter u_{piv} .*

Theorem 11.3 (Iterative refinement [9]). *Iterate $r_k = b - Mx_k$, $M\Delta x_k = r_k$, $x_{k+1} = x_k + \Delta x_k$. If $\kappa(M)\epsilon_{\text{mach}}$ is sufficiently smaller than 1, the iteration fixed-precision refinement (residuals in working precision) yields componentwise backward error $O(\epsilon_{\text{mach}})$; reducing the forward error to $O(\epsilon_{\text{mach}})$ requires residuals computed in extra precision.*

Proposition 11.4 (Refinement with a regularized factorization). *If refinement uses the factorization of $K + \Delta$ (regularized) but residuals $b - Kx_k$ of the true K , then $x_{k+1} - x^* = (K + \Delta)^{-1}\Delta(x_k - x^*)$, so the iteration converges iff $\rho((K + \Delta)^{-1}\Delta) < 1$. The regularization bias is thereby removed at linear rate.*

Definition 11.5 (Tolerance hierarchy). Numerical methods for these problems use tolerances of three kinds, ordered by the usual convention $\tau_{\text{int}} \gg \tau_{\text{feas}} \geq \tau_{\text{opt}} \gg \epsilon_{\text{mach}}$ (typical magnitudes in practice, not a derived result): a *feasibility* tolerance (primal residual and bound violation), an *optimality* tolerance (dual residual and gap), and an *integrality* tolerance τ_{int} . A *pivot* tolerance and a *drop* tolerance are set relative to ϵ_{mach} and to $\|B\|$ respectively.

Lemma 11.6 (Perturbation error). *Let costs be perturbed to $c + \varepsilon$, let $\tilde{x}$ be optimal for the perturbed problem and x^* optimal for the original. Then $0 \leq c^\top \tilde{x} - c^\top x^* \leq \varepsilon^\top (x^* - \tilde{x}) \leq \|\varepsilon\|_\infty (\|x^*\|_1 + \|\tilde{x}\|_1)$. The bound reduces to $\|\varepsilon\|_\infty \|x^*\|_1$ only for one-sided perturbations ($\varepsilon \geq 0$, $x \geq 0$). A cleanup phase with the original costs removes this error [19].*

12 Verification Mathematics

Let $(\hat{x}, \hat{y}, \hat{s})$ be a candidate for the *original* (unscaled, unpresolved) problem.

Definition 12.1 (Residuals).

$$\varepsilon_P = \frac{\|\text{viol}(A\hat{x} - b)\|_\infty + \max_j \text{viol}_j(l \leq \hat{x} \leq u)}{1 + \|\hat{b}\|_\infty}, \quad \varepsilon_D = \frac{\|\text{viol}(c + Q\hat{x} - A^\top \hat{y} - \hat{s})\|_\infty}{1 + \|c\|_\infty},$$

$$\varepsilon_G = \frac{\left| \frac{1}{2} \hat{x}^\top Q \hat{x} + c^\top \hat{x} - d(\hat{y}, \hat{s}, \hat{x}) \right|}{1 + \left| \frac{1}{2} \hat{x}^\top Q \hat{x} + c^\top \hat{x} \right| + |d(\hat{y}, \hat{s}, \hat{x})|},$$

where viol measures the violation of the row sense (zero when satisfied), d is the dual objective, and the dual variables satisfy the sign conditions ($\hat{s}_j \geq 0$ at a lower bound, ≤ 0 at an upper bound, and so on).

Definition 12.2 (Status classification). With thresholds $\tau_1 < \tau_2$: *optimal* if $\max\{\varepsilon_P, \varepsilon_D, \varepsilon_G\} \leq \tau_1$; *near-optimal* if $\leq \tau_2$; *inaccurate* otherwise; *infeasible* or *unbounded* if a certificate (Farkas system, unbounded ray) is verified.

Proposition 12.3 (Soundness of verifying on the original problem). *By Proposition 4.1 and the postsolve rules, residuals of the recovered point on the original data (A, b, c, Q, l, u) measure the actual optimality error, whereas residuals on a scaled or presolved problem may be smaller or larger by the scaling factors. Hence the classification must use the original data, and it is independent of which algorithm produced the point. If $\hat{x}$ and $(\hat{y}, \hat{s})$ are exactly feasible, weak duality gives objective $-z^* \in [0, |\text{gap}|]$; with nonzero residuals this holds only approximately, and a rigorous bound must scale $\varepsilon_P, \varepsilon_D$ by norms of the dual and primal solutions.*

Definition 12.4 (MILP verification). A candidate $\hat{x}$ is a certified MILP solution if $\varepsilon_P \leq \tau_{\text{feas}}$, bound violation $\leq \tau_{\text{feas}}$, and $\max_{j \in \mathcal{I}} |\hat{x}_j - \text{round}(\hat{x}_j)| \leq \tau_{\text{int}}$. Feasibility is checked on the reported point after any rounding of integer coordinates to their nearest integers, since rounding by up to τ_{int} can push rows beyond τ_{feas} . The reported quality is the gap defined in Section 3.4, not a KKT residual.

13 Scope and Limitations of the Theory

- MIQP, NLP and MINLP are not covered: they need branch-and-bound over QP relaxations or nonlinear theory not developed here.
- Nonconvex Q is not treated, and the methods here give no global optimality guarantee for it; inertia tests detect it only sometimes, since $X^{-1}S$ can mask negative curvature (Lemma 2.7 and the inertia theorem).
- Certificates: simplex termination gives a Farkas ray (Theorem 3.12); IPM and ADMM certificates rely on divergence behavior and are heuristic unless the ADMM theorem hypotheses are checked.
- First-order methods return solutions of moderate accuracy (10^{-4}); vertex accuracy needs crossover and simplex.
- MILP presolve, cut, branching and heuristic families here are a subset of those known in the literature; termination guarantees are for the exact tree search only, not under time or node limits.

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